CO1 – Security Features in Real-World Applications

1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:
The security feature/service required, such as confidentiality, integrity, authentication, authorization, availability

The security mechanism used to provide the identified security feature.

The security protocol or technology used to implement the mechanism.

Briefly explain how the mechanism/protocol provides the identified security feature.

Solution:

1. Introduction

Cryptography plays an important role in keeping information safe when we use applications and online services every day. It helps protect messages, passwords, payment details, and personal data from unauthorized people. Different security features such as confidentiality, integrity, authentication, authorization, and availability use security and cryptographic techniques in different ways. For example, encryption keeps information private, while authentication helps verify that a user is really who they claim to be. Technologies such as TLS, MFA, tokenization, and end-to-end encryption are commonly used in applications like Instagram, Telegram, Google Pay, Netflix, and YouTube. Thus, cryptography and cybersecurity help make online communication and transactions more secure, private, and trustworthy.

2. Security Features in Real-World Applications

Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
Instagram	Authentication	Multi-Factor Authentication (MFA)	TOTP / Authentication App	A user enters a password and a one-time code. This makes it harder for someone else to access the account, even if the password is known.
Telegram	Confidentiality	End-to-End Encryption	MTProto / Secret Chats	Messages are encrypted so that only the intended users can read them. This keeps private conversations confidential.
Google Pay	Confidentiality & Integrity	Encryption and Tokenization	TLS / Payment Tokenization	Sensitive payment information is protected using encryption, while tokenization replaces actual card details with a secure token. This helps protect payment data during transactions.
Netflix	Authorization	Access Control	OAuth 2.0 / Access Tokens	After logging in, access controls and tokens help determine what resources the user is allowed to access. This prevents unauthorized access to protected accounts and services.
YouTube	Availability	Distributed Servers and Load Balancing	CDN / HTTPS	Videos are delivered through multiple servers, helping the service remain available when many users access it at the same time. HTTPS also protects communication between the user and the service.

3. Related Explanations and Definitions

- **Cryptography:** The technique of protecting information by changing it into a form that unauthorized people cannot easily understand.
- **Encryption:** Converts readable data (plaintext) into unreadable data (ciphertext). A key is used to encrypt and decrypt the information.
- **Confidentiality:** Ensures that information can only be viewed by authorized people. Encryption is a common way to provide confidentiality.
- **Integrity:** Ensures that information is not changed or damaged without permission. Cryptographic techniques such as hashes, message authentication codes, and TLS help detect or prevent unwanted changes.
- **Authentication:** Checks whether a user is really who they claim to be. Passwords, MFA, and one-time passwords are common authentication methods.
- **Authorization:** Determines what an authenticated user is allowed to access or do. Access tokens and permissions are commonly used for this purpose.
- **Availability:** Ensures that a service or resource is available when users need it. CDNs, backups, replication, and load balancing help maintain availability.
- **TLS:** Transport Layer Security protects data exchanged between a device and a server by providing encryption and integrity protection.
- **End-to-End Encryption:** Encrypts information so that it can be read only by the intended communicating users or endpoints.
- **Tokenization:** Replaces sensitive information, such as a card number, with a token. The token can be used instead of exposing the original information.
- **MFA:** Multi-Factor Authentication requires two or more types of evidence to verify a user's identity, such as a password and a one-time code.
- **CDN:** A Content Delivery Network uses servers in different locations to deliver content efficiently and help keep online services available.

4. How Cryptography Supports Cybersecurity

- It protects sensitive information from unauthorized access.
- It helps secure communication between users and online services.
- It helps protect passwords, payment information, and private messages.
- It supports data integrity by helping detect unauthorized changes to information.
- It works together with authentication and authorization to provide safer access to applications.
- It builds trust in online communication, transactions, and digital services.

5. Key Learnings

1. Cryptography is a major part of modern cybersecurity.
2. Different applications use different security mechanisms depending on what they need to protect.
3. Encryption mainly helps provide confidentiality and can also support integrity when used with authentication mechanisms.
4. MFA provides stronger authentication by requiring more than just a password.
5. Tokenization reduces the exposure of sensitive payment information.
6. Authorization controls what an authenticated user is allowed to access.
7. Availability is mainly supported by infrastructure such as CDNs, replication, and load balancing rather than encryption alone.

8. Good cybersecurity usually combines cryptography, authentication, authorization, and reliable infrastructure rather than depending on one technique.

6. Conclusion

Cryptography and cybersecurity work together to protect the applications and services we use every day. From logging into Instagram to sending private messages on Telegram and making payments through Google Pay, security mechanisms are used to protect users and their information. Understanding confidentiality, integrity, authentication, authorization, and availability helps us understand how real-world applications stay secure and trustworthy.

2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Solution:

1. Introduction

Security controls are methods used by applications and online services to protect users, data, and systems from security attacks. These controls help achieve three important security goals: Confidentiality, Integrity, and Availability (CIA). They can prevent attacks before they happen, detect suspicious activity, and help recover after an attack or system failure. Common controls include encryption, authentication, access control, backups, monitoring, firewalls, and secure communication protocols.

2. Security Controls Used in Real-World Applications

Application	**Security Control**	**CIA**	**Mechanism / Protocol**	**How it helps**
WhatsApp	End-to-End Encryption	Confidentiality	Signal Protocol	Encrypts messages so that only the sender and intended receiver can read them. This helps prevent unauthorized people from reading private conversations.
Gmail	Authentication and Secure Email Communication	Confidentiality	MFA / TLS	MFA helps prevent unauthorized account access, while TLS protects email data while it is being transferred between systems.
Online Banking	Encryption and Transaction Verification	Confidentiality & Integrity	HTTPS / TLS / MFA	TLS protects banking information during communication, while MFA and transaction verification help prevent unauthorized transactions.
Amazon	Access Control and Secure Payments	Confidentiality & Integrity	HTTPS / TLS / MFA / Tokenization	Protects account and payment information and helps prevent unauthorized changes or purchases.
Google Drive	Access Control and Data Backup	Confidentiality & Availability	OAuth 2.0 / TLS / Replication	Access controls decide who can view or edit files. TLS protects data during transfer, and replication helps keep files available if a system fails.

3. Explanation of CIA

- **Confidentiality:** Keeps information private and prevents unauthorized people from seeing it. Example: WhatsApp encrypting private messages.
- **Integrity:** Makes sure information is not changed without permission. Example: TLS and transaction verification help protect data from unauthorized changes.

- **Availability:** Makes sure a service or data is available when users need it. Example: Google Drive uses replication and reliable infrastructure to keep files available.

4. Preventing, Detecting and Recovering from Attacks

Application	Prevent	Detect	Recover
WhatsApp	Encryption prevents unauthorized reading of messages.	Users can identify unknown linked devices or sessions.	Unknown devices can be removed and the account can be secured again.
Gmail	MFA and strong authentication prevent unauthorized login.	Security alerts and unusual-login detection can identify suspicious activity.	The user can change the password, remove unwanted sessions, and secure the account.
Online Banking	MFA and transaction verification help prevent unauthorized payments.	Banks monitor transactions for suspicious or unusual activity.	A compromised account or payment method can be blocked and investigated.
Amazon	MFA, TLS, and payment security reduce unauthorized access and fraud.	Unusual login and purchase activity can be detected.	Passwords or payment methods can be changed or removed and the account can be secured.
Google Drive	Access controls prevent unauthorized users from viewing or editing files.	Account activity and sharing activity can be monitored.	Deleted or lost data can be recovered using available backups or copies.

5. Important Mechanisms and Protocols

- **Encryption:** Converts readable data into protected data so unauthorized people cannot easily read it.
- **MFA:** Uses more than one method to verify a user, such as a password and a one-time code.
- **TLS:** Protects data while it travels between a user's device and a server.
- **HTTPS:** Uses HTTP together with TLS to provide secure web communication.
- **Signal Protocol:** Provides end-to-end encryption for secure messaging applications such as WhatsApp.
- **OAuth 2.0:** Allows applications to access resources using controlled access tokens instead of directly sharing passwords.
- **Tokenization:** Replaces sensitive payment information with a token to reduce exposure of the original data.
- **Access Control:** Determines who is allowed to view, change, or use a resource.
- **Replication / Backup:** Keeps additional copies of data or services so they can be restored or remain available after a failure.

6. Other Simple Real-World Applications

- WhatsApp – messaging and communication
- Gmail – email and communication
- Online Banking – financial transactions
- Amazon – online shopping and payments
- Google Drive – online file storage and sharing
- Google Pay – digital payments
- Instagram – social media and communication
- Netflix – online video streaming

7. Key Learnings

9. Security controls are used to protect users, data, and online services.
10. Confidentiality keeps information private.

11. Integrity helps ensure that information is not changed without permission.
12. Availability keeps applications and data accessible when users need them.
13. Encryption and TLS are important for protecting data during communication.
14. MFA provides stronger protection against unauthorized account access.
15. Access control makes sure users can only access the resources they are allowed to use.
16. Monitoring helps detect suspicious activity and possible attacks.
17. Backups, replication, and recovery methods help restore data or services after failures or attacks.
18. Real-world applications normally use several security controls together instead of relying on only one control.

8. Conclusion

Security controls are an important part of modern cybersecurity. Applications such as WhatsApp, Gmail, online banking, Amazon, and Google Drive use different controls to protect users and data. Encryption protects confidentiality, security checks help maintain integrity, and backups and distributed systems help maintain availability. Together, these controls help applications prevent, detect, and recover from security attacks.